

Project Specification: Pavan Koundinya Portfolio Website

Overview

Build a modern, responsive, space-themed personal portfolio website for a Software Development Engineer with experience in scalable backend systems, mission operations, data dissemination platforms, and space technology.

The website should feel premium, futuristic, and inspired by Mission Control / spacecraft operations rather than a generic sci-fi theme.

The site must be fully static and deployable via GitHub Pages.

Technical Requirements

Framework

- Next.js 15+
- TypeScript
- Tailwind CSS
- Framer Motion
- Static Export Compatible
- GitHub Pages Compatible

Deployment

Support deployment using GitHub Pages.

Requirements:

- Static export enabled
- No server-side rendering
- No backend APIs
- No database dependencies

Build command:

`npm run build`

Output:

`out/`

Design Language

Theme

Mission Control + Space Exploration

Avoid:

- Cartoon rockets
- Gaming aesthetics
- Neon cyberpunk overload

Target feel:

- NASA
- ESA
- SpaceX mission dashboard
- Satellite operations consoles
- Deep space exploration

Visual Style

Dark mode only.

Primary Background:

020617

Secondary Background:

0F172A

Primary Accent:

3B82F6

Secondary Accent:

8B5CF6

Text:

E2E8F0

Muted Text:

94A3B8

Success:

10B981

Global Effects

Starfield

Animated star background.

Requirements:

- Subtle movement
- Low CPU usage
- Visible on all sections
- Parallax effect

Animations

Use Framer Motion.

Requirements:

- Fade-in on scroll
 - Smooth page transitions
 - Hover interactions
 - No excessive animations
-

Navigation

Sticky top navigation.

Logo:

PK

Menu Items:

- Mission Brief
- Experience
- Projects

- Skills
- Mission Log
- Resume
- Contact

Desktop: Horizontal navbar

Mobile: Hamburger menu

Section 1: Hero

Full viewport height.

Background: Animated stars.

Content:

Name: Pavan Koundinya

Title: Software Development Engineer

Tagline:

Building scalable software systems on Earth while contributing to space technology.

Buttons:

Primary: Explore Mission

Secondary: Download Resume

Social Icons:

- GitHub
- LinkedIn
- Email

Visual Element:

Floating satellite/spacecraft wireframe animation on right side.

Section 2: Mission Brief

Equivalent to About Me.

Title:

Mission Brief

Content:

Computer Science Engineer and Software Development Engineer with experience designing scalable backend systems, distributed applications, data dissemination platforms, and mission-support software.

Passionate about solving complex engineering challenges and building reliable systems that operate at scale.

Include:

- Years of experience
- Current role
- Domain expertise

Cards:

Backend Engineering Distributed Systems Cloud Technologies Space Technology

Section 3: Experience

Title:

Mission Timeline

Vertical timeline component.

Timeline Entries:

Current Role

Company: Kaleideo

Role: Software Development Engineer 2

Include:

- Scalable backend systems
- Data dissemination platform
- Cross-functional collaboration
- Production support

Additional experience entries should be configurable through JSON.

Timeline animations:

Rocket launch progression effect.

Section 4: Projects

Title:

Mission Portfolio

Layout:

Responsive card grid.

Each project card contains:

- Name
- Description
- Technologies
- Impact
- GitHub link
- Demo link

Featured Projects Section

Large cards for:

1. Data Dissemination Platform

Description: Scalable platform for delivering mission and operational data.

1. Mission Operations Tools

Description: Software supporting spacecraft operations workflows.

1. Internal Engineering Platforms

Description: Tools improving engineering productivity.

Project cards should animate on hover.

Section 5: Skills Galaxy

Title:

Skills Galaxy

Visualization:

Skills grouped into constellations.

Categories:

Backend Systems

- Java
- Spring Boot
- Python
- REST APIs

Cloud & Infrastructure

- AWS
- Docker
- Kubernetes
- CI/CD

Databases

- PostgreSQL
- MySQL
- MongoDB

Frontend

- React
- Next.js
- TypeScript

Soft Skills

- Leadership
- Ownership
- Collaboration

Hovering a skill should display proficiency level.

Section 6: Mission Log

Title:

Mission Log

Purpose:

Professional achievements.

Display as mission records.

Examples:

Mission Record 001

Built scalable data dissemination platform supporting operational workflows.

Mission Record 002

Contributed to mission-support systems and engineering tooling.

Mission Record 003

Led implementation of backend services for critical applications.

Layout:

Terminal-style mission logs.

Section 7: Resume

Title:

Mission Documents

Content:

Short summary.

Button:

Download Resume PDF

Resume file location:

public/resume.pdf

Section 8: Contact

Title:

Open Communication Channel

Fields:

- Name

- Email
- Message

Since GitHub Pages is static:

Use mailto link.

Alternative:

Integrate Formspree.

Social Links:

- GitHub
 - LinkedIn
 - Email
-

Footer

Display:

Pavan Koundinya

Software Development Engineer

Built with Next.js and deployed via GitHub Pages.

Include social links.

Responsiveness

Must support:

- Desktop
- Laptop
- Tablet
- Mobile

Breakpoints:

320px+ 768px+ 1024px+ 1440px+

No horizontal scrolling.

Accessibility

Requirements:

- Keyboard navigation
 - Semantic HTML
 - Proper heading hierarchy
 - Alt text for graphics
 - WCAG-compliant contrast
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Performance

Target:

Lighthouse Score > 90

Requirements:

- Lazy loading
 - Optimized images
 - Minimal JavaScript
 - Fast first paint
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SEO

Metadata:

Title: Pavan Koundinya | Software Development Engineer

Description: Software Development Engineer specializing in scalable backend systems, distributed applications, and mission-support software.

Open Graph support.

Twitter card support.

Structured data support.

Content Management

All personal content should be stored in:

/src/data/

Files:

profile.ts experience.ts projects.ts skills.ts achievements.ts

So content can be updated without touching UI code.

Deliverables

Generate:

- Complete Next.js project
- TypeScript
- Tailwind configuration
- Framer Motion integration
- Reusable components
- GitHub Pages deployment workflow
- README with deployment instructions
- Sample content populated
- Responsive design
- Space-themed visual system
- Production-ready code